

Initial martingale sampling diagnostics

ARC-Challenge, ARC-Easy and CommonsenseQA | GPT-4o mini and DeepSeek V4 Pro

Prepared for an initial methodological review
17 September 2026

Purpose. This note consolidates the diagnostics currently implemented in ***[martingale_property_test_sampling.ipynb](#)***. It is intended as an early readout of the experiment, not as a final hypothesis test.

Main observations

- The average local residual magnitude is very small in five of the six runs. The exception is GPT-4o mini on CommonsenseQA ($1.79\text{e-}03$), about 838 times the DeepSeek value on the same selected questions.
- All observed current predictions are correct: there are no incorrect states in any run. The planned correct-versus-incorrect comparison therefore cannot yet be evaluated.
- The model probabilities are strongly saturated. Mean maximum probabilities range from 0.998364 to 1.000000. The current experiment mainly examines near-one-hot states rather than the full probability simplex.
- The signed class residual tests mostly support an average close to zero, but this is a necessary rather than sufficient condition for a martingale. Positive and negative local departures can cancel when averaged.

1. Experimental coverage

For each visited history, the experiment computes the conditional expected next probability vector and subtracts the current vector: $r_t = E[p_{t+1} | F_t] - p_t$. The saved tensors contain 49 transitions and five classes. Inference in the notebook first averages dependent steps within a trajectory and then resamples complete trajectories.

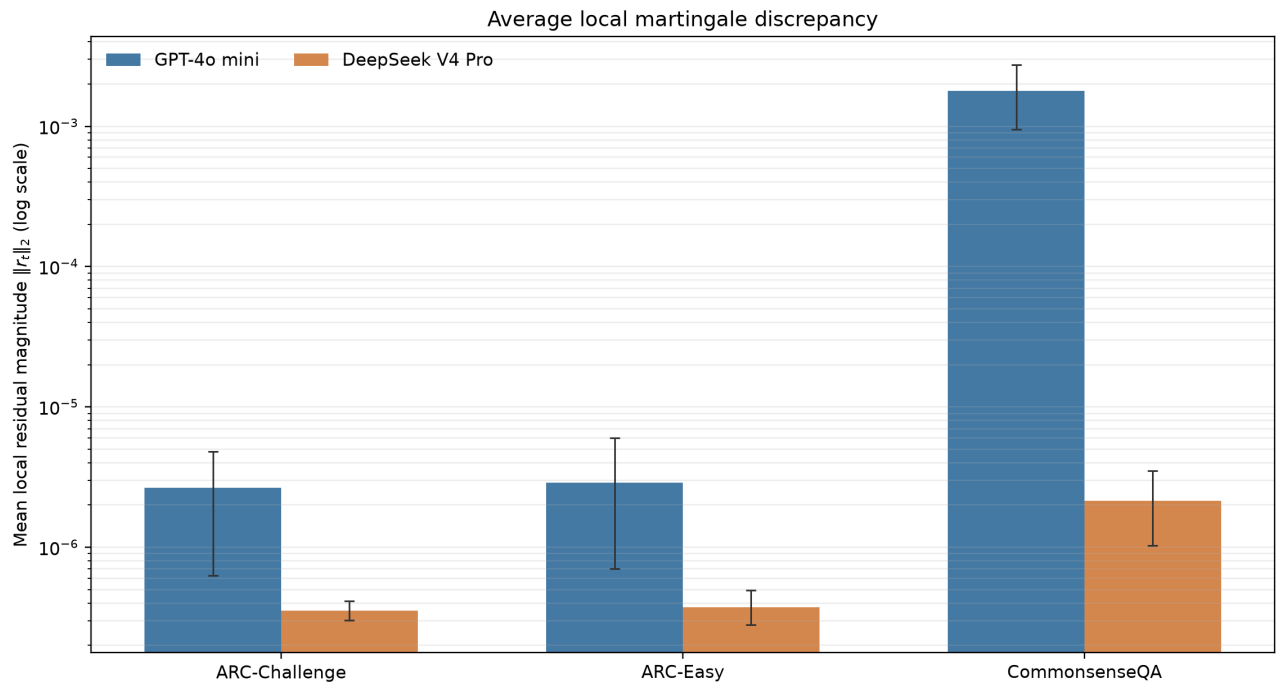
Dataset	Model	Questions	Traj./question	Clusters	States	Incorrect
ARC-Challenge	GPT-4o mini	5	5	25	1225	0
ARC-Challenge	DeepSeek V4 Pro	5	5	25	1225	0
ARC-Easy	GPT-4o mini	5	5	25	1225	0
ARC-Easy	DeepSeek V4 Pro	10	5	50	2450	0
CommonsenseQA	GPT-4o mini	5	5	25	1225	0
CommonsenseQA	DeepSeek V4 Pro	5	5	25	1225	0

The selected question indices match across models for ARC-Challenge (True) and CommonsenseQA (True), but not for ARC-Easy (False). ARC-Easy also uses 10 questions for DeepSeek and 5 for GPT-4o mini, so that model comparison is descriptive rather than paired.

Tests reproduced from the notebook

- Class-wise mean signed residual with a 95% percentile bootstrap interval, resampling complete trajectories.
- Two-sided one-sample t-test of trajectory-mean residuals against zero, with a Bonferroni threshold across five classes.
- L2 residual magnitude by step, including mean, median and pointwise trajectory-bootstrap intervals.
- Signed residual conditional on the current class-probability bin.
- Correct-versus-incorrect state comparison where both groups are observed.

2. Overall residual magnitude

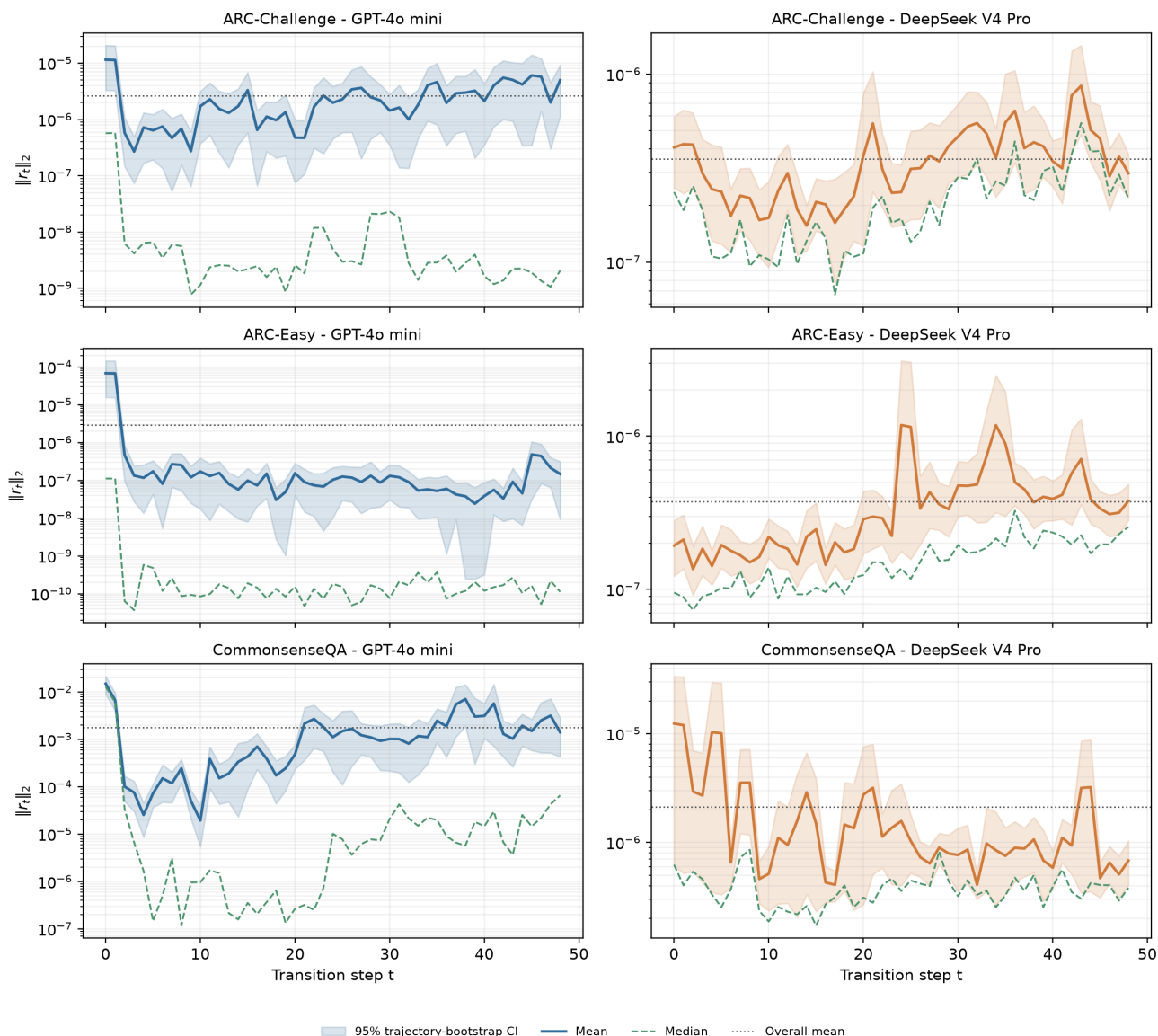


Dataset	Model	Mean L2	95% CI	Median	Maximum
ARC-Challenge	GPT-4o mini	2.656e-06	[6.31e-07, 4.80e-06]	2.91e-09	8.32e-05
ARC-Challenge	DeepSeek V4 Pro	3.539e-07	[3.02e-07, 4.12e-07]	1.86e-07	6.30e-06
ARC-Easy	GPT-4o mini	2.885e-06	[7.00e-07, 6.03e-06]	1.46e-10	8.03e-04
ARC-Easy	DeepSeek V4 Pro	3.737e-07	[2.81e-07, 4.93e-07]	1.38e-07	4.67e-05
CommonsenseQA	GPT-4o mini	1.789e-03	[9.49e-04, 2.75e-03]	6.08e-06	1.02e-01
CommonsenseQA	DeepSeek V4 Pro	2.135e-06	[1.03e-06, 3.51e-06]	3.49e-07	2.57e-04

The median is well below the mean in every run, especially for GPT-4o mini. This indicates a right-skewed distribution: most transitions have negligible discrepancy, with a small number of larger excursions driving the average. CommonsenseQA with GPT-4o mini stands apart both in its mean and its maximum residual.

3. Evolution over the trajectory

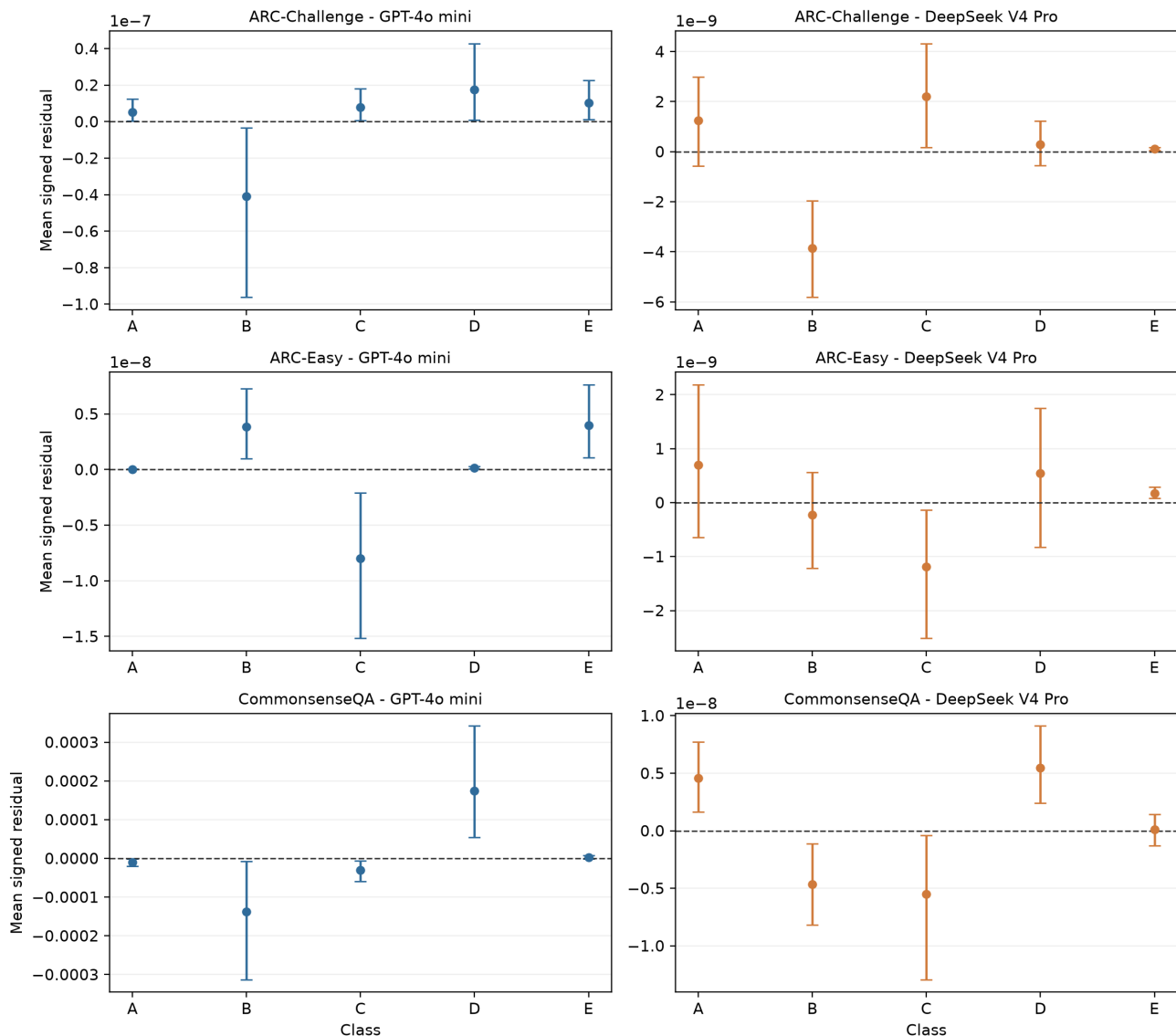
Residual magnitude over the 49 transitions



The log scale is intentional: the runs differ by several orders of magnitude. There is no common monotone increase across all six panels. Instead, the means are driven by intermittent step-specific spikes. The median often remains close to the numerical floor while the mean moves, which again points to a small subset of trajectories or histories accounting for most of the observed discrepancy.

4. Are mean signed residuals different from zero?

Class-wise mean signed residuals with 95% trajectory-bootstrap intervals



Across 30 dataset-model-class combinations, 23 percentile bootstrap interval(s) exclude zero and 6 t-test(s) remain significant after the within-run Bonferroni correction. The class residuals are compositional and sum to approximately zero at each state, so the five tests are not independent. The corrected result is included as a conservative screen, not as a complete joint test.

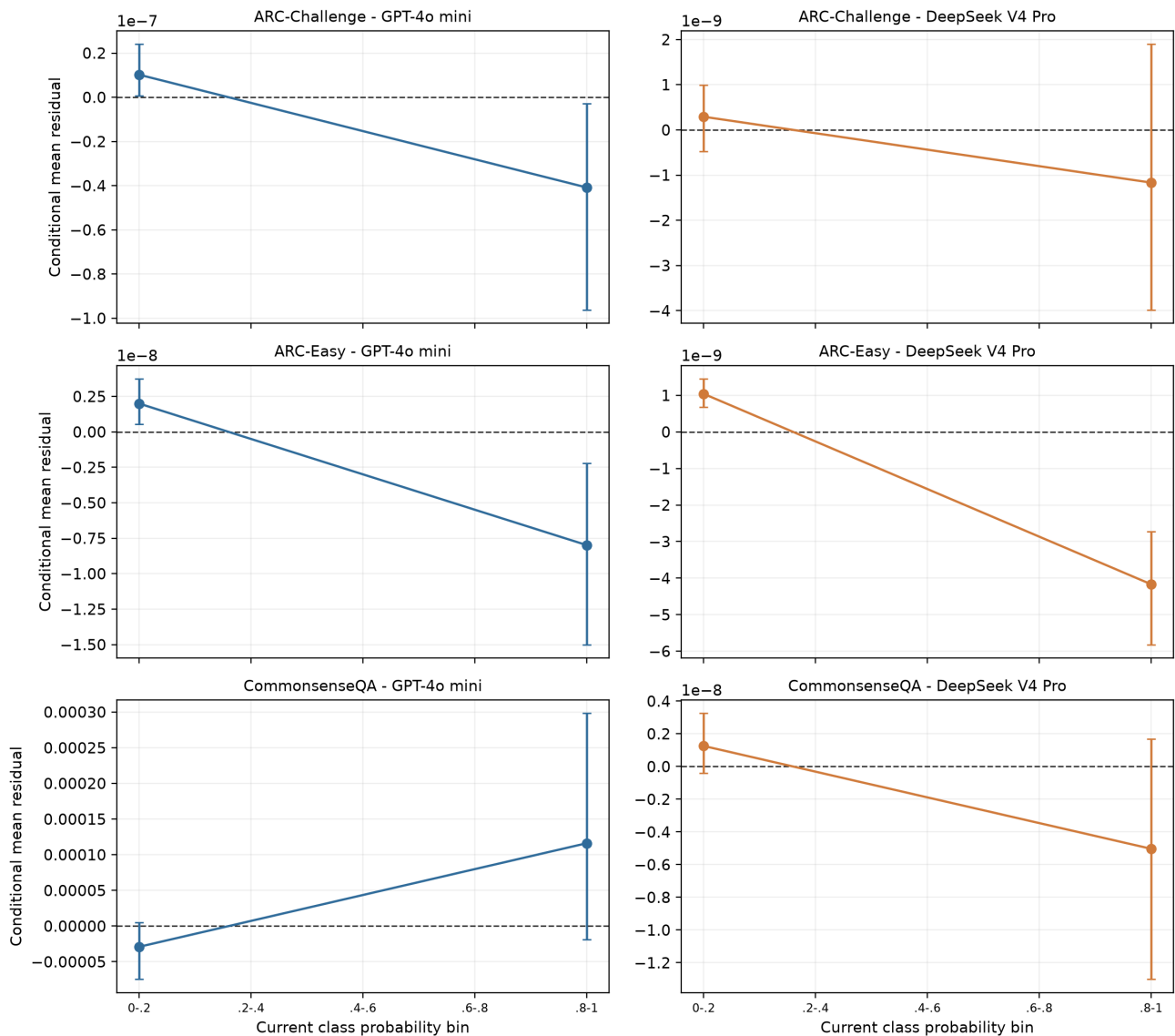
Classes retained by the Bonferroni screen

Dataset	Model	Class	Mean	Bootstrap CI	p-value	Bonf.
ARC-Challenge	DeepSeek V4 Pro	B	-3.857e-09	[-5.81e-09, -1.96e-09]	8.04e-04	yes
ARC-Challenge	DeepSeek V4 Pro	E	1.117e-10	[5.39e-11, 1.74e-10]	1.63e-03	yes
ARC-Easy	GPT-4o mini	A	2.035e-11	[8.84e-12, 3.43e-11]	5.62e-03	yes
ARC-Easy	DeepSeek V4 Pro	E	1.733e-10	[8.12e-11, 2.86e-10]	1.75e-03	yes
CommonsenseQA	DeepSeek V4 Pro	A	4.557e-09	[1.62e-09, 7.74e-09]	9.30e-03	yes
CommonsenseQA	DeepSeek V4 Pro	D	5.480e-09	[2.39e-09, 9.10e-09]	4.79e-03	yes

A mean of zero is only a necessary marginal implication of the martingale condition. It does not rule out conditional violations that reverse sign across histories. The L2 diagnostic and the probability-conditioned analysis are therefore important complements.

5. Residuals conditional on current probability

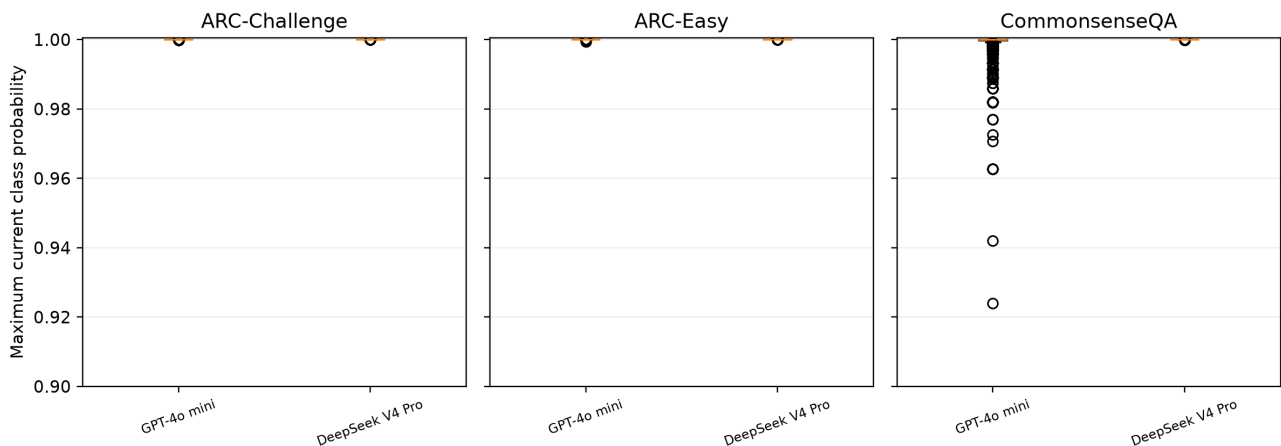
Signed residual conditional on current class probability



This diagnostic asks whether the expected next probability systematically moves up or down from its current level. In principle, a positive residual in the 0.8-1.0 bin would indicate self-reinforcement of already high beliefs. Here the interpretation is limited by the extreme concentration at the edges: the lowest bin contains almost all non-winning classes and the highest bin contains almost all winning classes, while middle bins often contain few or no observations. The plotted intervals should therefore be read together with the bin counts in the appendix.

6. Confidence saturation and correctness

Prediction confidence is concentrated close to one



Dataset	Model	Mean max p	Minimum max p	Mean entropy	Correct states
ARC-Challenge	GPT-4o mini	0.9999974	0.9999235	3.382e-05	1225/1225
ARC-Challenge	DeepSeek V4 Pro	0.9999997	0.9999950	4.767e-06	1225/1225
ARC-Easy	GPT-4o mini	0.9999988	0.9994174	1.221e-05	1225/1225
ARC-Easy	DeepSeek V4 Pro	0.9999997	0.9999605	4.760e-06	2450/2450
CommonsenseQA	GPT-4o mini	0.9983639	0.9239832	9.260e-03	1225/1225
CommonsenseQA	DeepSeek V4 Pro	0.9999990	0.9998171	1.320e-05	1225/1225

The requested correct-versus-incorrect comparison cannot be run because the incorrect group is empty in all six files. This is not evidence that martingale violations never occur in incorrect states; it means the current question samples do not contain such states. The near-one-hot probabilities also create a ceiling effect: once a winning class has probability close to one, both upward movement and the residual norm are mechanically constrained.

7. Dataset-level reading

ARC-Challenge

On matched question indices, GPT-4o mini has a mean L2 residual of $2.66\text{e-}06$ and DeepSeek $3.54\text{e-}07$ (ratio 7.5). Both remain very small in absolute terms and all evaluated states are correct.

ARC-Easy

Both models show very small discrepancies ($2.89\text{e-}06$ for GPT-4o mini and $3.74\text{e-}07$ for DeepSeek). However, the questions are not matched and DeepSeek uses twice as many questions, so the ratio should not be read as a controlled model comparison.

CommonsenseQA

GPT-4o mini has a mean L2 residual of $1.79\text{e-}03$, compared with $2.13\text{e-}06$ for DeepSeek (ratio 838). This is the clearest separation in the current data and occurs on matched question indices. GPT-4o mini is also less saturated here (minimum maximum probability 0.924), giving the residual more room to vary.

Provisional conclusion

Within the narrow set of sampled, almost perfectly confident states, the saved results are broadly compatible with small local martingale discrepancies. The strongest exception is GPT-4o mini on CommonsenseQA, where both the average and maximum L2 residuals are much larger than in the other runs. At this stage it is safer to describe that as a signal for follow-up rather than a model-level conclusion.

Recommended next run

- Increase the number of questions substantially and use identical question indices for both models on every dataset.
- Deliberately include difficult or previously misclassified questions so that incorrect states are represented.
- For population-level uncertainty, resample questions as the outer cluster and trajectories within questions, rather than treating all question-trajectory pairs as fully independent.
- Pre-specify a practically meaningful tolerance for the L2 residual; testing a non-negative norm against exactly zero is not informative with finite numerical precision.
- Inspect the largest CommonsenseQA GPT-4o mini residuals at the prompt/history level to determine whether they arise from a reproducible behavioral transition or from provider/log-probability truncation.

Appendix A. Full class-wise test results

Dataset	Model	Class	Mean residual	Bootstrap 95% CI	t-test p	Bonf.
ARC-Challenge	GPT-4o mini	A	5.273e-09	[4.27e-10, 1.23e-08]	1.13e-01	no
ARC-Challenge	GPT-4o mini	B	-4.086e-08	[-9.61e-08, -3.41e-09]	1.11e-01	no
ARC-Challenge	GPT-4o mini	C	7.734e-09	[7.30e-10, 1.80e-08]	1.12e-01	no
ARC-Challenge	GPT-4o mini	D	1.752e-08	[9.46e-10, 4.28e-08]	1.32e-01	no
ARC-Challenge	GPT-4o mini	E	1.033e-08	[1.06e-09, 2.26e-08]	8.17e-02	no
ARC-Challenge	DeepSeek V4 Pro	A	1.254e-09	[-5.73e-10, 2.98e-09]	1.88e-01	no
ARC-Challenge	DeepSeek V4 Pro	B	-3.857e-09	[-5.81e-09, -1.96e-09]	8.04e-04	yes
ARC-Challenge	DeepSeek V4 Pro	C	2.201e-09	[1.79e-10, 4.32e-09]	5.38e-02	no
ARC-Challenge	DeepSeek V4 Pro	D	2.909e-10	[-5.59e-10, 1.23e-09]	5.42e-01	no
ARC-Challenge	DeepSeek V4 Pro	E	1.117e-10	[5.39e-11, 1.74e-10]	1.63e-03	yes
ARC-Easy	GPT-4o mini	A	2.035e-11	[8.84e-12, 3.43e-11]	5.62e-03	yes
ARC-Easy	GPT-4o mini	B	3.852e-09	[9.69e-10, 7.27e-09]	2.73e-02	no
ARC-Easy	GPT-4o mini	C	-7.988e-09	[-1.52e-08, -2.10e-09]	2.80e-02	no
ARC-Easy	GPT-4o mini	D	1.543e-10	[3.94e-11, 2.84e-10]	2.27e-02	no
ARC-Easy	GPT-4o mini	E	3.962e-09	[1.05e-09, 7.61e-09]	3.13e-02	no
ARC-Easy	DeepSeek V4 Pro	A	6.976e-10	[-6.48e-10, 2.18e-09]	3.41e-01	no
ARC-Easy	DeepSeek V4 Pro	B	-2.285e-10	[-1.21e-09, 5.64e-10]	6.22e-01	no
ARC-Easy	DeepSeek V4 Pro	C	-1.188e-09	[-2.51e-09, -1.40e-10]	5.93e-02	no
ARC-Easy	DeepSeek V4 Pro	D	5.455e-10	[-8.21e-10, 1.75e-09]	4.16e-01	no
ARC-Easy	DeepSeek V4 Pro	E	1.733e-10	[8.12e-11, 2.86e-10]	1.75e-03	yes
CommonsenseQA	GPT-4o mini	A	-9.525e-06	[-2.02e-05, -1.33e-06]	6.89e-02	no
CommonsenseQA	GPT-4o mini	B	-1.372e-04	[-3.13e-04, -8.14e-06]	1.03e-01	no
CommonsenseQA	GPT-4o mini	C	-3.070e-05	[-5.90e-05, -6.63e-06]	3.12e-02	no
CommonsenseQA	GPT-4o mini	D	1.747e-04	[5.43e-05, 3.42e-04]	3.40e-02	no
CommonsenseQA	GPT-4o mini	E	2.754e-06	[-1.55e-06, 7.43e-06]	2.55e-01	no
CommonsenseQA	DeepSeek V4 Pro	A	4.557e-09	[1.62e-09, 7.74e-09]	9.30e-03	yes
CommonsenseQA	DeepSeek V4 Pro	B	-4.633e-09	[-8.17e-09, -1.14e-09]	1.88e-02	no
CommonsenseQA	DeepSeek V4 Pro	C	-5.498e-09	[-1.29e-08, -4.10e-10]	1.12e-01	no
CommonsenseQA	DeepSeek V4 Pro	D	5.480e-09	[2.39e-09, 9.10e-09]	4.79e-03	yes
CommonsenseQA	DeepSeek V4 Pro	E	9.395e-11	[-1.28e-09, 1.43e-09]	8.93e-01	no

The p-values are two-sided. Because each returned p-value already combines both tails, it is compared with $\alpha = 0.05$ (or $0.05/5$ for Bonferroni), not $\alpha/2$. The $\alpha/2$ split appears in the endpoints of the corresponding two-sided confidence interval.

Appendix B. Probability-bin counts

Dataset	Model	[0,.2)	[.2,.4)	[.4,.6)	[.6,.8)	[.8,1]
ARC-Challenge	GPT-4o mini	4900	0	0	0	1225
ARC-Challenge	DeepSeek V4 Pro	4900	0	0	0	1225
ARC-Easy	GPT-4o mini	4900	0	0	0	1225
ARC-Easy	DeepSeek V4 Pro	9800	0	0	0	2450
CommonsenseQA	GPT-4o mini	4900	0	0	0	1225
CommonsenseQA	DeepSeek V4 Pro	4900	0	0	0	1225

Counts pool classes and transitions for the descriptive conditional analysis. They are not independent sample sizes; dependence is handled by resampling complete trajectories when constructing the plotted intervals.

Scope and reproducibility

Source files: the six *martingale_sampling_results.npz* artifacts under *data/results_mp_sampling*. Analysis follows the current sampling notebook with 20,000 bootstrap draws for class-wise and overall summaries and 10,000 draws for step and probability-bin plots. Random seed: 42. This document was generated directly from the saved arrays; no model calls were repeated.